A One-Line-Latency Vision Enhancement Terminal

PYNQ-Z2 | reading aid for the low-vision | 2026 Embedded Chip & System Design Competition, AMD track, self-selected topic (junior group) | Zynq-7000 XC7Z020 | Vivado / Vitis HLS 2026.1

Motivation

Around 17 million people in China live with low vision, and shop-bought electronic magnifiers are expensive while phone/PC pipelines take 100-300 ms from capture to display - enough latency to cause dizziness. For this class of device the key metric is not frame rate but end-to-end latency. This project therefore puts the whole enhancement chain inside the pixel stream, with no frame buffer at all.

Architecture

Main path: PS frame -> VDMA -> enhance -> pixel_unpack -> HDMI OUT, with the kernel sitting in the live pixel stream; measured 21 pixel clocks on the board, seven presets switched while video keeps running.
The frame source is the VDMA because this board's HDMI output only works at 640x480 (800x600/720p/1080p consume no frames; the stock bitstream behaves identically - the pins are unbuffered), while the HDMI input IP cannot run at 640x480. With no common mode, frames come from DDR and the kernel still processes real pixels.
The enhance kernel runs local contrast -> tone curve -> binarise/invert -> split screen, all integer, one pixel per clock; 59 configuration bits are driven from the PS at run time.
Verification path: PS <-> DDR <-> AXI DMA <-> operator, readable back, so bit-exact checks need no display.
Three rotating line buffers plus per-row 3-tap shift registers give each buffer one read and one write per pixel, which is what fits II = 1 into a dual-port BRAM.

Operators

enhance: 3 BRAM, 2 DSP, 2168 FF, 1659 LUT (3%), II = 1, estimated Fmax 194.59 MHz at 7 ns. The operator family behind it - Sobel, 3x3 Gaussian, 3x3 sharpen - is hand-written HLS too (II = 1, 194.6 MHz, under 1.1k LUT each), and colour-space conversion reuses PYNQ's verified color_convert IP instead of being rewritten.

Correctness

enhance passes C simulation against an independently written C++ golden model in two configurations (contrast+split, and tone+binarise+invert+mono): 9216 beats each, 0 errors. Through the AXI-DMA path, all eight presets are compared channel by channel with the numpy model: 0 mismatches out of 2,764,800 channels each, at 15.6 ms per frame. A second pair of counters brackets the kernel and stamps the first beat in and out: the measured pipeline latency is 16 pixel clocks, i.e. 160 ns at 100 MHz - about 1% of a 720p60 line, which is what having no frame buffer buys. Border policy (horizontal replicate, vertical line-buffer state) is defined once and shared by hardware, testbench and Python model, which is why the comparison can use zero tolerance.

What Made It Work - and What Did Not

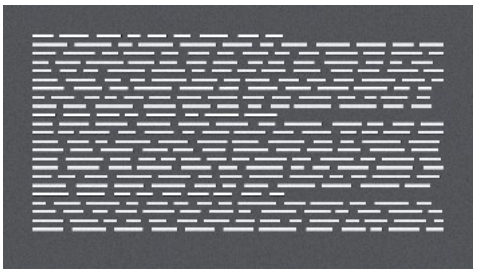
A free-running kernel (ap_ctrl_none) never drains its pipeline when the input stops, so an AXI-DMA transfer can never close; the DMA path therefore uses a bounded variant (ap_ctrl_hs plus a pixel count). The AXI DMA buffer-length register is 14 bits by default, i.e. 16 KB - not enough for a frame, so it was widened to 23 bits. The PYNQ base overlay is a 4448-line Vivado 2024.1 script; rebuilding it with 2026.1 needs exactly three patches.

Reusable PYNQ Skills and Reproducibility

Four board-independent skills ship with the project: overlay handoff linting (.hwh), bit-exact image comparison, measurement collection with PS-vs-PL table generation, and AXI-DMA transfer helpers with cache maintenance and timeouts. Each documents scope, usage, failure modes and what was verified. Everything is scripted, so a clean machine rebuilds both bitstreams with two commands.

Operator Family on the DMA Path (same board, same 1280x720 image)

Measured on the board	PS (Cortex-A9, numpy)	PL (Zynq + DMA)
enhance, 8 presets	-	15.6 ms/frame, 0/2764800 wrong
Pipeline latency	-	16 pixel clocks = 160 ns
Sobel family, frame time	**3108.4 ms**	**15.8 ms (196.7x)**



Sobel output captured from the PL path (DMA)

Tool versions: Vivado / Vitis HLS 2026.1, PYNQ 3.1.1 image, device xc7z020clg400-1. Repository: README.md, docs/ (docs/hdmi_status.md records the HDMI measurements and the two hardware limits), report/design_report.md, report/comparison.md.